

STUDENT GRADE MANAGEMENT SYSTEM

README

1. Project Overview

The Student Grade Management System is a simple Python-based project designed to manage basic academic information of students. It allows the user to manage student details, subjects, marks, grades, attendance, assignments, results, teachers, and overall performance. The project uses basic Python concepts and separate modules, making it easy to understand and suitable for academic use.

2. Main Features

- Student details management – Add, view, update, delete, and search students.
- Subject management – Maintain subject names, codes, credits, and teachers.
- Marks management – Store internal, midterm, and final marks and calculate totals.
- Grade management – Calculate grades according to percentage.
- Attendance management – Record attendance and calculate attendance percentage.
- Teacher management – Store basic teacher information.
- Result management – Calculate percentage and pass/fail status.
- Assignment management – Track assignments, marks, and submission status.
- Performance management – Calculate overall academic performance.

3. Project Modules

Module	Purpose
1.MAIN_PAGE.py	Main menu and dashboard.
student_info.py	Student personal and academic details.
subject_info.py	Subjects, codes, credits, and teachers.
marks_info.py	Marks and total calculation.
grade_info.py	Grade calculation from percentage.
attendance_info.py	Attendance records and percentage.
teacher_info.py	Teacher information.
result_info.py	Results and pass/fail status.
assignment_info.py	Assignments and submission status.
performance_info.py	Overall performance calculation.

4. Technologies Used

- Python 3
- VS Code or any Python-supported IDE
- Basic Python concepts: lists, functions, loops, conditions, input/output, and modules

5. Project Structure

The project is divided into different Python files. Each file handles one part of the system, which keeps the code simple and easier to understand.

```
1.MAIN_PAGE.py
student_info.py
subject_info.py
marks_info.py
grade_info.py
attendance_info.py
teacher_info.py
result_info.py
assignment_info.py
performance_info.py
```

6. How to Run the Project

1. Install Python 3.
2. Open the project folder in VS Code.
3. Keep all Python files in the same folder.
4. Run the main file:

```
python 1.MAIN_PAGE.py
```

5. Use the menu shown on the screen to access the modules.

7. Data Storage

This version uses simple Python lists to store data while the program is running. It does not use a database or external files. Therefore, the entered data is cleared when the program is closed. This keeps the project simple and beginner-friendly.

8. Grading System

Percentage	Grade
90% and above	A+
80% – 89%	A
70% – 79%	B+
60% – 69%	B
50% – 59%	C
40% – 49%	D
Below 40%	F

9. Performance Calculation

The overall performance score combines average marks, attendance, and assignment performance.

Overall Performance = (Average Marks × 60%) + (Attendance × 20%) + (Assignment Average × 20%)

10. Testing

The project can be tested by entering sample students, subjects, marks, attendance, and assignments. Each module should be checked using its available insert, view, update, delete, search, and calculation

options.

11. Limitations

- Data is stored only while the program is running.
- No database is used.
- No login or authentication system is included.
- The interface is command-line based.

12. Future Improvements

- Add MySQL for permanent storage.
- Create a Tkinter graphical interface.
- Add login and authentication.
- Generate student report cards.
- Add charts and detailed reports.
- Add backup and restore options.

13. Conclusion

The Student Grade Management System is a beginner-friendly Python project that demonstrates how a larger application can be divided into smaller modules. It provides practical experience with functions, lists, conditions, loops, calculations, and modular programming while solving a simple academic management problem.